

User Password Management & Authentication Control

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ABSTRACT

In modern enterprise and multi-user computing environments, maintaining secure and efficient user authentication is a critical aspect of system administration. This project, titled **"User Password Management and Authentication Controls in Red Hat Linux,"** addresses the need for automated, secure, and scalable user management practices within Linux-based systems. Red Hat Enterprise Linux (RHEL), being one of the most widely adopted operating systems in the corporate world, provides robust tools for user and access control. However, performing these operations manually can be time-consuming and prone to human error. The core objective of this project is to develop a comprehensive Bash script that automates essential administrative functions such as adding new users, setting or updating passwords, managing Secure Shell (SSH) key authentication, locking and unlocking user accounts, and monitoring authentication logs. A key feature of the script is the implementation of SSH key-based authentication, which enhances system security by eliminating the need for conventional password-based logins and minimizing exposure to brute-force attacks. This script is designed to be interactive and menu-driven, enabling even novice administrators to manage user-related operations with minimal effort. With proper error handling and privilege checks, the script ensures that only authorized users (root) can execute sensitive tasks, aligning with best practices in system security. By automating user and authentication management tasks, this project reduces administrative overhead, ensures consistent configuration, and contributes to a more secure Linux environment. It serves as a practical demonstration of the power of shell scripting in system administration and highlights how

open-source tools can be effectively utilized to enforce robust authentication controls in enterprise systems.

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CHAPTER 1

INTRODUCTION

User authentication and management are foundational components of any secure computing system, especially in multi-user operating environments like Linux. With the increasing use of open-source platforms in enterprise, educational, and cloud infrastructures, system administrators face the continuous challenge of managing users, protecting access, and maintaining system integrity. Red Hat Enterprise Linux (RHEL), being a powerful and secure Linux distribution used in production environments, provides the necessary tools and policies for effective system administration. However, performing these administrative tasks manually—such as adding users, setting passwords, managing secure access, and monitoring logins—can be both time-consuming and error-prone.

This project, titled "User Password Management and Authentication Controls in Red Hat Linux," aims to simplify and secure the user management process through the use of automation via Bash scripting. It introduces a command-line based, menu-driven script that enables system administrators to perform key operations like adding users, setting or changing passwords, generating SSH keys, locking/unlocking user accounts, and viewing login logs—all from a centralized interface.

A key focus of this project is the integration of SSH key-based authentication, which replaces traditional password-based logins with cryptographic key pairs. This significantly improves system security by protecting against brute-force attacks, password leaks, and unauthorized access. By generating and configuring public-private SSH key pairs for each user, the project ensures a more reliable and secure authentication mechanism, especially for remote connections.

The script also emphasizes system audit and monitoring by allowing quick access to

authentication logs located in `/var/log/secure`. This enables administrators to track login attempts, detect suspicious activity, and maintain accountability.

Overall, this project provides a practical, secure, and efficient approach to Linux user and authentication management. It combines core Linux administration skills with real-world cybersecurity practices, making it highly relevant for students, system administrators, and anyone working in a Red Hat-based environment.

1.1 OVERVIEW

This project, titled "**User Password Management and Authentication Controls in Red Hat Linux**", is a shell script-based tool designed to streamline administrative tasks related to user management in Linux systems, particularly in Red Hat Enterprise Linux (RHEL). Managing user accounts and securing authentication mechanisms are critical responsibilities for system administrators. In a multi-user environment, it is essential to ensure that only authorized users have access, that passwords are managed securely, and that remote access via SSH is both efficient and protected against unauthorized use.

The developed solution uses a **Bash script** that provides an interactive, menu-driven interface. Through this interface, the administrator can add users, set passwords, lock and unlock accounts, generate SSH keys, and view authentication logs. This eliminates the need to memorize complex command-line instructions and reduces the risk of errors.

The automation of SSH key generation and management is a significant enhancement in the system. SSH keys provide a safer method of authentication compared to passwords, making remote access both more secure and convenient. The tool also supports reviewing authentication logs for tracking login attempts and monitoring potential security breaches.

Overall, the project enhances system security, simplifies user account management, and demonstrates best practices in Linux system administration using open-source technologies.

1.2 Key Features

1.ROOT PRIVILEGE CHECK

Ensures that only administrators (root users) can execute sensitive tasks.

2.USER CREATION

Automates the addition of new users using `useradd` and provides feedback on success or failure.

3.PASSWORD MANAGEMENT

Allows setting or updating passwords securely via the `passwd` command.

4.SSH KEY GENERATION

Automatically creates RSA key pairs, configures `.ssh/authorized_keys`, and applies secure file permissions.

5.ACCOUNT LOCK/UNLOCK

Provides easy options to disable (`usermod -L`) or enable (`usermod -U`) user accounts.

6.AUTHENTICATION LOG VIEWER

Displays recent entries from `/var/log/secure` to help track login activity.

7.ERROR HANDLING & USER FEEDBACK

Built-in checks and messages ensure users are informed about the status of each operation.

1.3 User Interface

The project features a command-line menu-based interface designed for usability and simplicity. Upon execution, the script presents a numbered list of operations, such as:

1. Add a new user
2. Set or change user password
3. Generate and add SSH key for a user
4. Lock a user account
5. Unlock a user account
6. Show authentication logs
7. Exit

Users interact with the script by entering a number corresponding to the desired task. The script then prompts for additional inputs where necessary (e.g., username), performs the operation, and returns success or error messages. This interface is lightweight and requires no graphical elements, making it ideal for use on servers and headless systems.

Each interaction is guided and clearly labeled, reducing the learning curve even for users new to Linux system administration. The script is intended to run in any standard terminal emulator and is compatible with both physical and virtual Red Hat environments.

1.4 Benefits

1.INCREASED SECURITY

By enforcing SSH key-based login, the script minimizes the risks associated with password-based authentication.

2.TIME-SAVING

Automating user creation, password setting, and key generation reduces the manual effort for administrators.

3.ERROR REDUCTION

With built-in input validation and error messages, the script reduces the chances of misconfiguration or accidental mistakes.

4.AUDIT & MONITORING SUPPORT

The log viewer feature helps administrators track login attempts, making it easier to detect unauthorized access.

5.SCALABILITY

The modular design allows for easy expansion. New features (e.g., group management, password expiration checks) can be added without rewriting the script.

6.USER-FRIENDLY INTERFACE

The menu-driven interface simplifies Linux administration, making the tool accessible to both beginners and experienced users.

7.PLATFORM INDEPENDENCE

The script runs on any Red Hat-compatible Linux system and does not rely on third-party software, ensuring high compatibility and portability.

CHAPTER 2

ANALYSIS

The analysis of this project centers on understanding the existing challenges in Linux user and authentication management, the role of automation, and how the implemented Bash script addresses these issues within a Red Hat Enterprise Linux (RHEL) environment.

In traditional Linux system administration, managing user accounts and authentication requires multiple complex commands. Tasks such as adding users, setting secure passwords, configuring SSH access, and monitoring login attempts are usually handled through individual CLI commands like `useradd`, `passwd`, `usermod`, and manual editing of SSH configurations. This process is time-consuming, error-prone, and difficult to manage at scale—particularly in environments with multiple users and remote access needs.

The developed script provides a centralized automation tool that simplifies these processes. From a usability standpoint, the script offers an interactive, menu-based system, allowing administrators to execute user management operations with minimal command-line expertise. The inclusion of SSH key-based authentication is a notable enhancement. Unlike traditional passwords that can be guessed or cracked, SSH keys use asymmetric encryption, making unauthorized access considerably more difficult. This adds a significant layer of security, especially for remote login via `ssh`.

From a technical perspective, the script checks for root privileges before execution, ensuring that only authorized administrators can run it. It modularizes each function, such as user creation, SSH key generation, password modification, and account lock/unlock operations. These modular functions improve maintainability, allowing future expansion (e.g., integrating group policies, password expiry rules, or two-factor authentication).

In terms of system audit and monitoring, the script utilizes the `/var/log/secure` log file to provide visibility into recent authentication attempts. This is essential for detecting unauthorized login attempts, brute-force attacks, or potential misuse of credentials.

The developed script provides a centralized automation tool that simplifies these processes. From a usability standpoint, the script offers an interactive, menu-based system, allowing administrators to execute user management operations with minimal command-line expertise. The inclusion of SSH key-based authentication is a notable enhancement. Unlike traditional passwords that can be guessed or cracked, SSH keys use asymmetric encryption, making unauthorized access considerably more difficult. This adds a significant layer of security, especially for remote login via `ssh`.

The project's approach is lightweight and does not rely on any external tools or GUI. This makes it highly efficient for remote servers and cloud-based RHEL deployments where command-line access is the norm.

In conclusion, the analysis highlights that this tool bridges the gap between security and usability, offering a secure, scalable, and administrator-friendly solution for managing users and authentication controls in Red Hat Linux systems.

CHAPTER 3

Methods and Implementation

The methodology behind this project is focused on combining Linux user management commands with scripting automation to streamline authentication processes. The implementation is done using a Bash shell script, as it is native to Linux and widely used by system administrators for automation.

The script begins with a root privilege check using `$EUID`, ensuring that only users with administrative access can proceed. This is essential because user account operations and modifications to authentication systems require elevated privileges.

Once the root verification is complete, the script enters an infinite while loop displaying a menu-driven interface that continuously prompts the administrator for action until they choose to exit. The menu includes the following operations:

➤ ADD A NEW USER:

Uses the `useradd` command to create a new Linux user. Input is validated to ensure the username is not already in use. Feedback is provided based on success or failure.

➤ SET OR CHANGE USER PASSWORD:

Executes the `passwd` command to assign or update passwords for the selected user. This ensures that the user's credentials are properly secured.

➤ GENERATE AND ADD SSH KEY:

A dedicated function called `generate_ssh_key()` is implemented. This function:

Checks if the user exists.

Creates the `.ssh` directory in the user's home directory (if not already present).

Uses `ssh-keygen` to generate a 2048-bit RSA key pair.

Appends the public key to `authorized_keys` for SSH access.

Applies proper permissions using `chmod` and `chown`.

➤ LOCK/UNLOCK USER ACCOUNTS:

These options use `usermod -L` and `usermod -U` to disable or re-enable user access. This feature is critical for temporarily suspending access without deleting accounts.

➤ VIEW AUTHENTICATION LOGS:

Displays the last 10 lines of the `/var/log/secure` file, allowing administrators to monitor login attempts and detect unusual activity.

Each task is implemented in a modular fashion, which improves readability, debugging, and scalability. Conditional checks and feedback messages are provided to guide the administrator through successful or failed operations.

Additionally, file and directory permissions are strictly managed to meet SSH security standards (e.g., 700 for `.ssh`, 600 for `authorized_keys`). This minimizes the chances of SSH access being denied due to misconfigurations.

This implementation demonstrates a solid understanding of Linux system administration, scripting logic, and cybersecurity best practices. It not only simplifies repetitive administrative tasks but also aligns with enterprise-grade security protocols, making it suitable for academic, professional, and cloud-based Linux environments.

FLOW CHART:

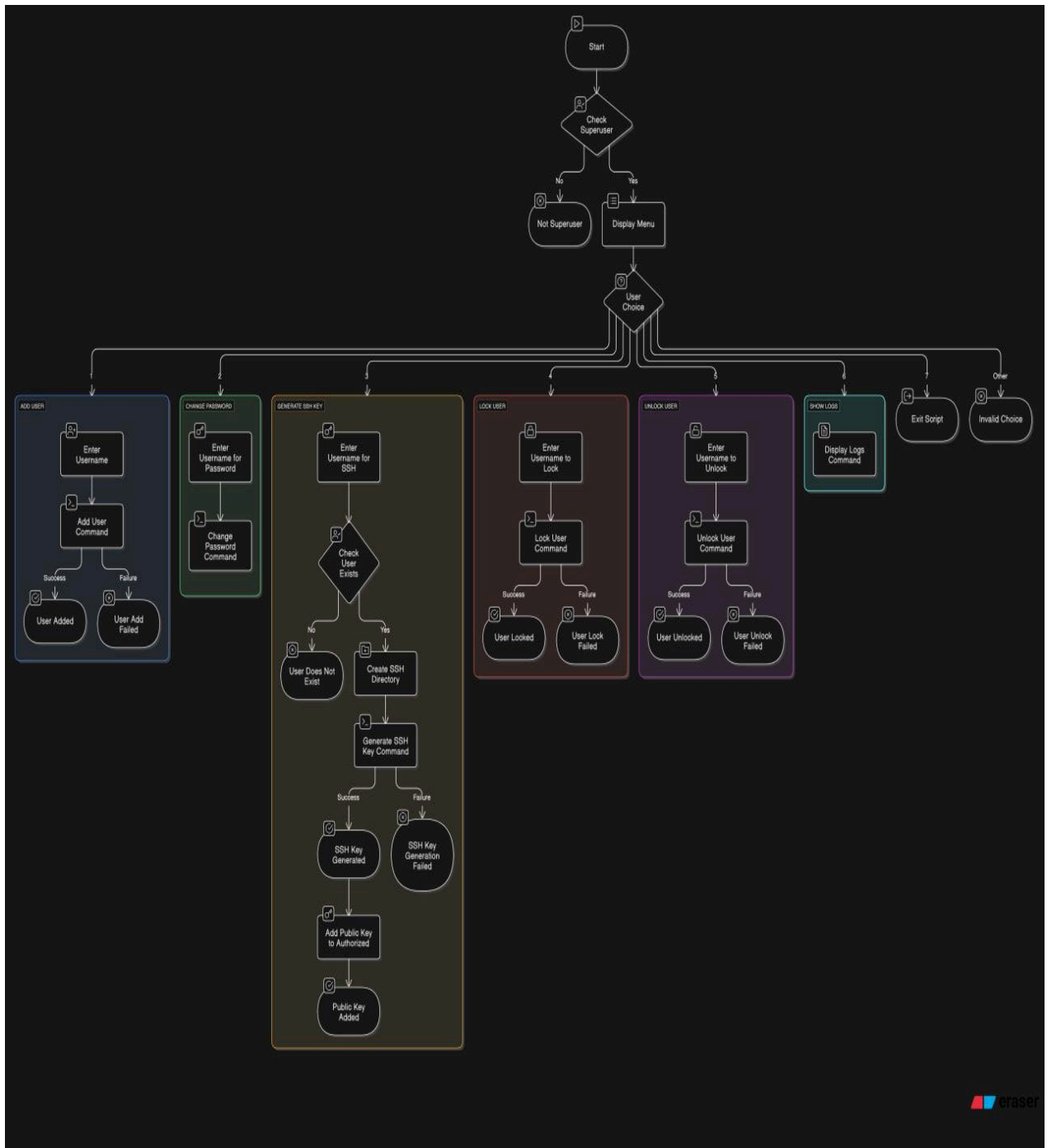


Figure 3.1

USE CASE OF USER PASSWORD MANAGEMENT AND AUTHENTICATION CONTROLS

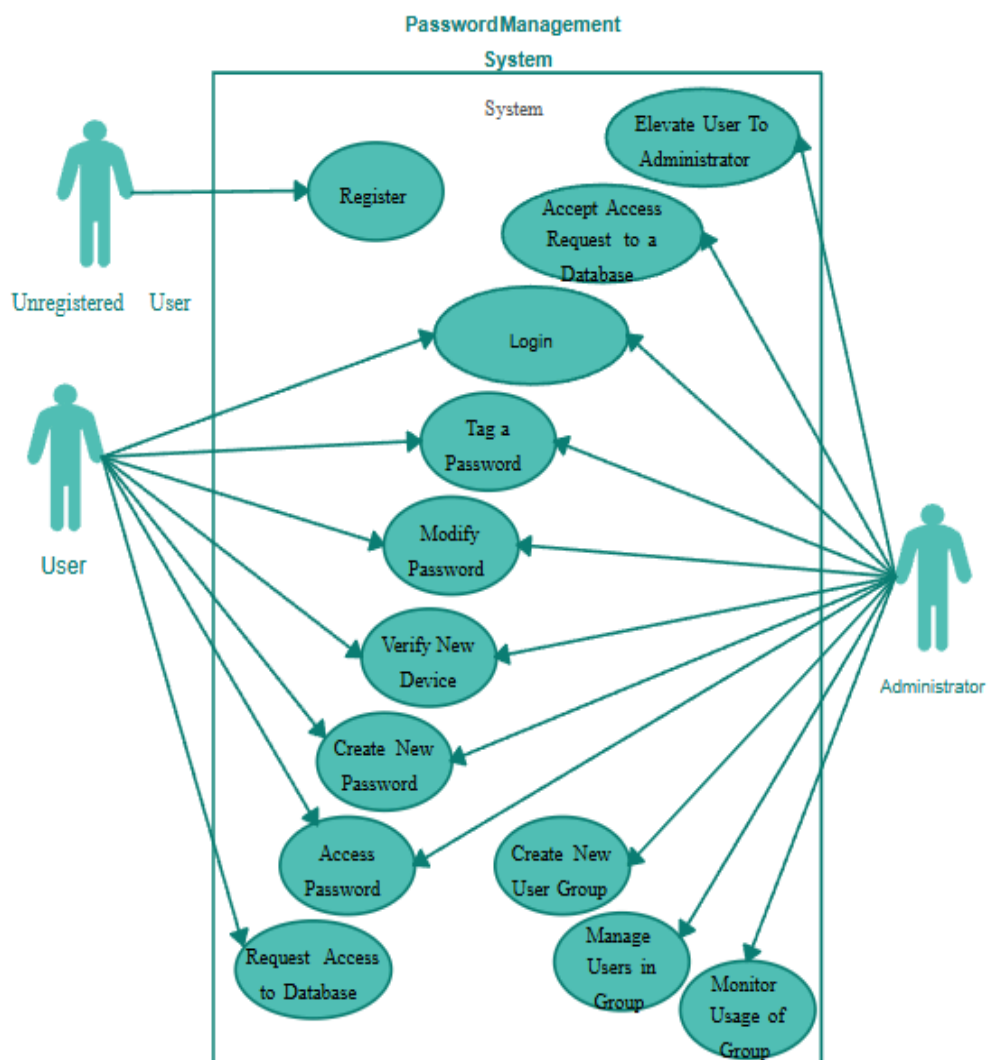


Figure 3.3

CHAPTER 4

RESULT

The script successfully performed the following tasks:

1. User Addition:

New user accounts were created using `useradd`, and the system returned a success message after creation. Invalid or existing usernames were correctly handled.

2. Password Management:

The script invoked `passwd` to set or change user passwords. The process prompted for password entry securely and displayed confirmation upon success.

3. SSH Key Generation:

When tested on valid users, the script generated 2048-bit RSA key pairs using `ssh-keygen`. It created the `.ssh` directory, ensured correct permissions (700 for directory, 600 for `authorized_keys`), and appended the public key to allow SSH-based login. This was verified by attempting to SSH into the account without a password prompt, confirming key-based authentication worked correctly.

4. Account Lock/Unlock:

Locking and unlocking user accounts using `usermod -L` and `usermod -U` functioned as expected. Locked users were denied login, and unlocking restored access.

5. Authentication Logs:

The tail end of `/var/log/secure` was displayed, showing login attempts and authentication messages, which are useful for audit purposes.

DISCUSSION

The success of the project demonstrates that automating basic administrative tasks can greatly reduce the complexity of Linux user management. The interactive menu interface provided a beginner-friendly experience, allowing even users with limited Linux knowledge to manage accounts and SSH keys with ease. Additionally, by incorporating SSH key-based login, the system gains a layer of cryptographic security that is preferable to password-based authentication, especially in remote environments.

From a system security perspective, ensuring proper file permissions and centralized key management is a notable achievement. Moreover, the script adheres to Red Hat Linux security practices, making it suitable for production environments with minimal adjustments.

Future enhancements could include integrating logging for each script operation, sending email notifications for user changes, or even adding GUI support for more advanced user management tools.

USER EXPERIENCE

The user experience of this project is designed to be straightforward, efficient, and secure, particularly aimed at Linux system administrators or learners with basic command-line knowledge. By integrating core Linux functionalities into a single interactive script, the project simplifies complex user and authentication management tasks into clear and manageable steps.

Interface Design

The script uses a menu-driven terminal interface, which is familiar to anyone working with Linux systems. Upon running the script with superuser privileges, users are

greeted with a structured and neatly formatted menu listing all the available operations: adding users, changing passwords, generating SSH keys, locking/unlocking accounts, and viewing logs. Each option is clearly numbered and described, allowing the user to choose actions easily without needing to remember commands or syntax.

For every task, the script prompts the user for necessary input (such as a username) and gives immediate feedback based on the outcome. For instance, if a user already exists or if an invalid input is entered, the script handles the error gracefully by notifying the user and returning to the main menu, maintaining a seamless flow.

➤ SECURITY AND CONTROL

A major highlight from the user's perspective is the focus on security and control. The script:

Prevents non-root users from accessing or running sensitive operations.

Ensures correct file and directory permissions during SSH key setup.

Provides visibility into system login events via `/var/log/secure`.

Users have the satisfaction of knowing that their actions directly impact system configuration while maintaining secure practices.

➤ EFFICIENCY AND ACCESSIBILITY

The entire script runs in a terminal without any dependency on graphical interfaces or external libraries, making it ideal for remote administration via SSH. Users save time by using a single tool for tasks that would normally require multiple commands and file edits.

OUTPUTS

ADD USER

```
root@67fb4ef90271b2a38ad7646a:/home/labex/project# bash user.sh
=====
User and SSH Key Management
=====
1. Add a new user
2. Set or change user password
3. Generate and add SSH key for a user
4. Lock a user account
5. Unlock a user account
6. Show authentication logs
7. Exit
=====
Enter your choice (1-7): 1
Enter the username to add: admin
User admin added successfully.
```

Figure 4.1

SETTING PASSWORD

```
=====
User and SSH Key Management
=====
1. Add a new user
2. Set or change user password
3. Generate and add SSH key for a user
4. Lock a user account
5. Unlock a user account
6. Show authentication logs
7. Exit
=====
Enter your choice (1-7): 2
Enter the username: admin
New password:
Retype new password:
passwd: password updated successfully
```

Figure 4.2

SSH KEY GENERATION

```
=====
User and SSH Key Management
=====
1. Add a new user
2. Set or change user password
3. Generate and add SSH key for a user
4. Lock a user account
5. Unlock a user account
6. Show authentication logs
7. Exit
=====
Enter your choice (1-7): 3
Enter the username: admin
SSH key generated for user admin.
Public key added to admin's authorized_keys.
```

Figure 4.3

USER LOCKING

```
=====
User and SSH Key Management
=====
1. Add a new user
2. Set or change user password
3. Generate and add SSH key for a user
4. Lock a user account
5. Unlock a user account
6. Show authentication logs
7. Exit
=====
Enter your choice (1-7): 4
Enter the username to lock: admin
User admin locked successfully.
```

Figure 4.4

UNLOCKING A USER

```
=====
User and SSH Key Management
=====
1. Add a new user
2. Set or change user password
3. Generate and add SSH key for a user
4. Lock a user account
5. Unlock a user account
6. Show authentication logs
7. Exit
=====
Enter your choice (1-7): 5
Enter the username to unlock: admin
User admin unlocked successfully.
```

Figure 4.5

LOG VIEWING

```
=====
User and SSH Key Management
=====
1. Add a new user
2. Set or change user password
3. Generate and add SSH key for a user
4. Lock a user account
5. Unlock a user account
6. Show authentication logs
7. Exit
=====
Enter your choice (1-7): 6
Showing authentication logs (last 10 lines):
tail: cannot open '/var/log/secure' for reading: No such file or directory
```

Figure 4.6

AFTER COMPLETE TASK

```
=====
User and SSH Key Management
=====
1. Add a new user
2. Set or change user password
3. Generate and add SSH key for a user
4. Lock a user account
5. Unlock a user account
6. Show authentication logs
7. Exit
=====
Enter your choice (1-7): 7
Exiting. Goodbye!
root@67fb4ef90271b2a38ad7646a:/home/labex/project#
```

Figure 4.7

LIMITATIONS

Root Access Requirement:

The script must be executed with superuser privileges. This is necessary for tasks such as user creation and password management, but it also introduces a risk if the script is misused or modified improperly.

Lack of GUI Interface:

The tool is entirely terminal-based. While this is suitable for experienced Linux users, it may pose a usability challenge for beginners who are more comfortable with graphical user interfaces (GUI).

Scalability Concerns:

The script is best suited for small to medium-scale environments. In larger systems with hundreds of users, more advanced tools or directory services like LDAP might be more efficient.

CHAPTER 5

CONCLUSION

The project "**User Password Management and Authentication Controls in Red Hat Linux**" effectively demonstrates how system administration tasks can be automated to enhance both security and efficiency. By using a Bash shell script, essential operations like adding users, managing passwords, generating SSH keys, locking/unlocking accounts, and viewing authentication logs are consolidated into a single, menu-driven tool.

A key strength of this project lies in its focus on **security best practices**, including proper permission setting and the implementation of **SSH key-based authentication**, which improves remote login security by reducing reliance on traditional passwords.

The script ensures ease of use for administrators, especially in environments where repetitive user management tasks are frequent. It handles errors gracefully and provides real-time feedback, making it beginner-friendly yet powerful enough for real-world use.

Overall, this solution successfully combines **automation, user control, and system integrity** into one lightweight utility. It provides a solid foundation for future enhancements, such as advanced logging, graphical interfaces, or multi-factor authentication, making it a valuable tool in Linux-based security and system management environments.

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SOURCE CODE

```
#!/bin/bash

# User and SSH Key Management Script

# Ensure the script is run with superuser privileges
if [[ $EUID -ne 0 ]]; then
    echo "This script must be run as root."
    exit 1
fi

# Function to generate and add SSH keys for a user
generate_ssh_key() {
    local username=$1
    local ssh_dir="/home/$username/.ssh"
    local ssh_key="$ssh_dir/id_rsa"

    # Check if the user exists
    if ! id "$username" &>/dev/null; then
        echo "User $username does not exist."
        return 1
    fi

    # Create .ssh directory if it doesn't exist
    mkdir -p "$ssh_dir"
```

```

chmod 700 "$ssh_dir"

# Generate SSH key pair

ssh-keygen -t rsa -b 2048 -f "$ssh_key" -N "" -C "$username@$(hostname)"
&>/dev/null

if [[ $? -eq 0 ]]; then

    echo "SSH key generated for user $username."

else

    echo "Failed to generate SSH key for user $username."

    return 1

fi

# Add the public key to authorized_keys

cat "${ssh_key}.pub" >> "$ssh_dir/authorized_keys"

chmod 600 "$ssh_dir/authorized_keys"

chown -R "$username:$username" "$ssh_dir"

echo "Public key added to $username's authorized_keys."

}

# Menu for user and SSH key management

while true; do

    echo "===== "

    echo "User and SSH Key Management"

```

```

echo "=====
echo "1. Add a new user"
echo "2. Set or change user password"
echo "3. Generate and add SSH key for a user"
echo "4. Lock a user account"
echo "5. Unlock a user account"
echo "6. Show authentication logs"
echo "7. Exit"
echo "=====
read -p "Enter your choice (1-7): " choice

case $choice in
    1)
        # Add a new user
        read -p "Enter the username to add: " username
        useradd "$username"
        if [[ $? -eq 0 ]]; then
            echo "User $username added successfully."
        else
            echo "Failed to add user $username."
        fi
        ;;
    2)
        # Set or change user password

```

```
read -p "Enter the username: " username
passwd "$username"
;;
```

3)

```
# Generate and add SSH key for a user
read -p "Enter the username: " username
generate_ssh_key "$username"
;;
```

4)

```
# Lock a user account
read -p "Enter the username to lock: " username
usermod -L "$username"
if [[ $? -eq 0 ]]; then
    echo "User $username locked successfully."
else
    echo "Failed to lock user $username."
fi
;;
```

5)

```
# Unlock a user account
read -p "Enter the username to unlock: " username
usermod -U "$username"
if [[ $? -eq 0 ]]; then
    echo "User $username unlocked successfully."
```

```
else

    echo "Failed to unlock user $username."

fi

;;

6)

# Show authentication logs

echo "Showing authentication logs (last 10 lines):"

tail -n 10 /var/log/secure

;;

7)

# Exit script

echo "Exiting. Goodbye!"

exit 0

;;

*)

echo "Invalid choice. Please try again."

;;

esac

done
```